

Jharkhand University of Technology, Ranchi**B. Tech. 2nd Semester Examination, 2024 (NEP)****Subject : Elements of Electronics Engineering****Subject Code : ESEL2****Time Allowed : 3 Hours****Full Marks : 70**

*Candidates are required to give their answer in their own words as far as practicable.**The figures in the right-hand margin indicate full marks.**Answer any five questions in which Question No.1 is compulsory.*

1. Choose the correct alternative:**2×7=14**

- (i) With rise in temperature the electrical conductivity of intrinsic semiconductor
 - (a) increases.
 - (b) decreases.
 - (c) first increases and then decreases.
 - (d) first decreases and then increases.
- (ii) Doping a semiconductor with a pentavalent impurity increases
 - (a) electron concentration.
 - (b) hole concentration.
 - (c) both electron and hole concentrations.
 - (d) None of these
- (iii) The capacitance of a reverse-biased p-n junction diode
 - (a) decreases with increasing reverse bias voltage.
 - (b) increases with increasing reverse bias voltage.
 - (c) remains constant regardless of reverse bias voltage.
 - (d) is zero.
- (iv) Which configuration of a Bipolar Junction Transistor (BJT) provides current gain but no voltage gain?
 - (a) Common Base (CB)
 - (b) Common Emitter (BE)
 - (c) Common Collector (CC)
 - (d) All configurations provide both current and voltage gain
- (v) In a Field-Effect Transistor (FET), the current between the source and drain terminals is controlled by
 - (a) injected minority carriers.
 - (b) applied gate-source voltage.
 - (c) base current.
 - (d) collector current.
- (vi) The ability of a Zener diode voltage regulator to maintain a constant output voltage despite changes in the load current is referred to as
 - (a) Line regulation.
 - (b) Load regulation.
 - (c) Ripple rejection.
 - (d) Frequency response.

- (vii) In a SCR, once triggered into conduction, the device remains conducting until
- (a) the anode current is reduced below the holding current.
 - (b) the gate current is reduced below the holding current.
 - (c) the gate current is reduced to zero.
 - (d) the anode voltage is reduced to zero.
2. (a) State and derive the mass action law for semiconductor. Discuss how the carrier concentration in semiconductors varies with temperature.
- (b) Distinguish conductor, insulator and semiconductor on the basis of energy band diagram. 7+7
3. (a) Compare LED, Laser Diode, and Photodiode based on their operational principles, characteristics, and applications.
- (b) Explain the operation and characteristics of diode circuits used for clipping, clamping with appropriate diagram. 7+7
4. (a) Explain the operation of a square wave generator using the 555 timer IC.
- (b) Explain the operation of a P-N junction diode in forward and reverse bias conditions with circuit diagram. Explain its V-I characteristics. 7+7
5. (a) Describe the working principle of a TRIAC with V-I characteristics. List the applications of a TRIAC.
- (b) Describe the working principle of a PIN diode with Circuit diagram. List the applications of a PIN diode. 7+7
6. (a) Explain the structure of a MOSFET with a neat diagram and describe its application as a switch.
- (b) Sketch a circuit diagram for the common emitter configuration of a transistor and explain its input and output characteristics.
7. Write short notes on *any two*: 7×2=14
- (a) Tunnel diode
 - (b) Unijunction Transistor (UJT)
 - (c) Intrinsic and Extrinsic Semiconductors
 - (d) SCR
-

Jharkhand University of Technology, Ranchi

B. Tech. 2nd Semester Examination, 2024 (NEP)

Subject : Engineering Mathematics-II

Subject Code : BSM02

Time Allowed : 3 Hours

Full Marks : 70

*Candidates are required to give their answers in their own words
as far as practicable.*

The figures in the margin indicate full marks.

Answer any five questions in which Question No. 1 is compulsory.

1. Choose the correct alternative:

2×7=14

- (i) The particular integral of $(D^2 + 4)y = \sin 3x$ is
 (a) $\frac{-\sin 3x}{5}$ (b) $\frac{-\sin 3x}{4}$
 (c) $\frac{\cos 3x}{4}$ (d) $\frac{\cos 3x}{5}$
- (ii) Using method of undetermined co-efficient y_p of $y'' + 4y' + 4y = 2x^2$ is of the form
 (a) $a + b\bar{x}$ (b) $a + bx + cx^2$
 (c) $ax + bx^2 + cx^3$ (d) None of these
- (iii) The differential equation $(x^2 - ay)dx = (ax - y^2)dy$ is
 (a) Homogeneous (b) Linear
 (c) Non-linear (d) Exact
- (iv) If $f(x) = \cos x$ in $(-\pi, \pi)$, then the Fourier series of $f(x)$ contains
 (a) only sine terms. (b) only cosine terms.
 (c) Both (a) and (b) (d) None of these
- (v) The nature of the differential equation $4\frac{\partial^2 u}{\partial x^2} + 4\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0$ is
 (a) Elliptic (b) Parabolic
 (c) Hyperbolic (d) None of these
- (vi) The odds in favor of an event A are 5 to 4. The probability of success of A is
 (a) $4/5$ (b) $5/9$
 (c) $4/9$ (d) 1
- (vii) If the mean of a Poisson distribution is m , then S.D. of this distribution is
 (a) m^2 (b) m
 (c) $\sqrt{m}$ (d) None of these

2. (a) Solve : $(x+1)\frac{dy}{dx} - y = e^{3x}(x+1)^2$ 4
 (b) Solve : $xydx - (x^2 + 2y^2)dy = 0$ 5
 (c) Obtain the complete solution of $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = e^{3x} + \sin 2x$. 5
3. (a) Using method of variation of parameters, solve $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = \frac{e^{3x}}{x^2}$. 7
 (b) Solve : $x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} - 4y = x^4$ 7
4. (a) Find the orthogonal trajectories of the family $y^2 = 4ax$. 7
 (b) A body originally at 80°C cools down to 60°C in 20 minutes, the temperature of the air being 40°C . What will be the temperature of the body after 40 minutes from the original? 7
5. (a) Find the half range sine series for $f(x) = x \cos x$ in $(0, \pi)$. 7
 (b) Solve : $p \tan x + q \tan y = \tan z$ 7
6. (a) Using the method of separation of variables, solve $3\frac{\partial u}{\partial x} + 2\frac{\partial u}{\partial y} = 0, u(x, 0) = 4e^{-x}$. 7
 (b) A tightly stretched string with fixed end points $x = 0$ and $x = l$ is initially at rest in its equilibrium position. If each of its points is given a velocity $\lambda x(l-x)$, find the displacement of the string at any distance x from one end at any time t . 7
7. (a) X is a continuous random variable with probability density function given by

$$\begin{aligned}
 f(x) &= kx & (0 \leq x < 2) \\
 &= 2k & (2 \leq x < 4) \\
 &= -kx + 6k & (4 \leq x < 6)
 \end{aligned}$$

Find k , mean and variance of X . 4

- (b) A certain screw making machine produces on average of 2 defective screws out of 100 and packs them in boxes of 500. Find the probability that a box contain 15 defective screws. 5
- (c) The random variable X is normally distributed with mean equal to 9 and standard deviation 3. Find $P(X \geq 15)$ and $P(0 \leq X \leq 9)$. $\{\phi(0 \leq Z \leq 2) = 0.4772, \phi(0 \leq Z \leq 3) = 0.4987\}$. 5

PCEL2
EE/EEE/ECE/CS/IT Cybersecurity/Data Sc./Comp. Engg.

Jharkhand University of Technology, Ranchi

B. Tech. 2nd Semester Examination, 2024 (NEP)

Subject : Fundamentals of Measurement and Sensors

Subject Code : PCEL2

Time Allowed : 3 Hours

Full Marks : 70

Attempt only five questions. Question No. 1 is compulsory.

All questions carry equal marks.

Assume suitable data and/or assumptions if required.

1. Multiple choice questions:

2×7=14

- (i) Scale of PMMC instrument is
 - (a) Exponential
 - (b) Linear
 - (c) Uniform
 - (d) Squared
- (ii) Which of the following instruments indicate the instantaneous value of the electrical quantity being measured at the time at which it is being measured?
 - (a) Absolute
 - (b) Indicating
 - (c) Recording
 - (d) Integrating
- (iii) For measurement of inductance having high value, we should use
 - (a) Hay's bridge
 - (b) Maxwell's bridge
 - (c) Maxwell Wein bridge
 - (d) None of these
- (iv) The shunt resistance R_{sh} used in Q-meter has a value of
 - (a) 0.02Ω to 0.04Ω
 - (b) 2Ω to 4Ω
 - (c) 0.2Ω to 0.4Ω
 - (d) $2 K \Omega$ to $4 K \Omega$
- (v) Piezoelectric transducer is used for measuring
 - (a) non-electrical quantities
 - (b) electrical quantities
 - (c) chemical quantities
 - (d) any quantity
- (vi) Piezoelectric accelerometers:
 - (a) should not be used for high frequencies above 100Hz
 - (b) should be used for low frequencies
 - (c) should use a monitoring source of low input impedance
 - (d) have a low natural frequency
- (vii) In strain gauge torque meter, the strain gauges should be mounted at an angle of _____ to the shaft axis.
 - (a) 45 degree
 - (b) 60 degree
 - (c) 0 degree
 - (d) 90 degree

PCEL2

2. Describe the constructional details of an Electrodynamic type wattmeter. Derive the expression for torque when the instrument is used on a.c. Explain why it is necessary to make the potential coil circuit purely resistive? (2)

4+6+4

3. (a) State the limitations of a Wheatstone bridge. How is it overcome?

6+8

- (b) An ac bridge shown in below figure has the following constants:

Arm AB — $C_1 = 0.5 \mu F$, $R_1 = 1 k\Omega$

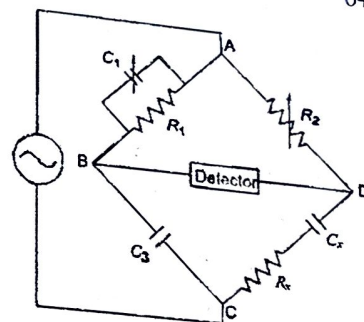
Arm AD — $R_2 = 2 k\Omega$

Arm BC — $C_3 = 0.5 \mu F$

Arm CD — C_x and R_x

Frequency — 1 kHz

Determine the unknown capacitance (C_x) and dissipation factor.



4. (a) Describe the construction, principle of working and applications of Hall Effect transducers.
- (b) A Hall effect transducer is used for the measurement of a magnetic field of 0.5 Wb/m^2 . The 2mm thick slab is made of Bismuth for which the Hall's co-efficient is $-1 \times 10^{-6} \text{ Vm/ (A-Wb m}^{-2})$ and the current is 3 A.

10+4

5. Explain the operation of seismic transducer for displacement mode measurements with its dynamic response curve using suitable equations and diagrams.

14

6. Discuss the operation of Strain gauge torque meter with diagram. Also, explain about the measurement of torque using this method with proper circuit diagram.

7+7

7. Write short notes on any two of the following:

7+7=14

(a) Absolute and Secondary Instruments

(b) Hay's bridge

(c) Three phase wattmeter

(d) LVDT

Jharkhand University of Technology, Ranchi**B. Tech. 2nd Semester Examination, 2024 (NEP)****Subject : Engineering Chemistry****Subject Code : BSC02****Time Allowed : 3 Hours****Full Marks : 70**

*Candidates are required to give their answer in their own words
as far as practicable.*

Answer any five questions in which Question No. 1 is compulsory.

The figures in the margin indicate full marks.

1. All the questions are multiple choices with one correct answer. Write the correct answer. 2×7=14

(i) A Solar cell is a _____ type of energy source.

(a) non-renewable

(b) renewable

(c) decomposable

(d) All of these

(ii) Which of the following is an example of QA?

(a) Verification

(b) Documentation

(c) Both (a) and (b)

(d) None of these

(iii) Which of the electrolyte used in Li-ion battery?

(a) Lead dioxide

(b) Lithium-based gel

(c) Sulphur dioxide

(d) Cobalt

(iv) Which one of the following is an example for top-down approach?

(a) Ball Milling Techniques

(b) Sol-gel process

(c) Both (a) and (b)

(d) None of these

(v) The metal which is protected by its own oxide—

(a) Cu

(b) Fe

(c) Au

(d) Al

(vi) Identification of functional groups in a compound can be established by using

(a) Chromatography

(b) IR Spectroscopy

(c) Both (a) and (b)

(d) None of these

(vii) The polymeric nanoparticles come under _____ dimensional nanomaterials.

(a) zero

(b) one

(c) two

(d) three

BSC02

(2)

92

2. (a) How do you classify materials for Engineering use? Describe in detail the role of Engineering Materials in our present world. 3+4
(b) Define and explain Quality Control and Quality Assurance in engineering contexts. Write difference between Qualitative and Quantitative analysis in Analytical Chemistry. 3+4
3. (a) What types of molecules are IR active? Explain why CO₂ molecule is IR active although it has Zero dipole moment. What is the importance of IR spectroscopy in finger print region? 2+2+3
(b) What is Molecular spectroscopy? How it differs from atomic spectroscopy? Differentiate between UV and IR Spectra of an Organic Compound. 2+2+3
4. (a) Define and explain Laws of Dry Corrosion. Discuss in detail the various methods of corrosion control with chemical equations. 3+4
(b) Discuss the factors influencing rate of corrosion. Differentiate between wet and dry corrosion. 3+4
5. (a) Explain principle and construction of lithium ion battery. Distinguish between lithium ion battery and sodium ion battery. 3+4
(b) Write difference between cell and battery. Write working principle and types of Fuel Cell. 3+4
6. (a) Define and explain Nanomaterials. How can classify Nanomaterials into different categories depending on the number of dimensions. 3+4
(b) Write application of Nanomaterials in various engineering field. 7
7. Write short notes on *any four* of the following: 3½×4=14
 - (a) Waterline Corrosion
 - (b) Solid Oxide Fuel Cell
 - (c) Alkali Fuel Cell
 - (d) Green Hydrogen Technology
 - (e) Supercapacitors

Jharkhand University of Technology, Ranchi

B. Tech. 2nd Semester Examination, 2024 (NEP)

Subject : Biology for Engineers

Subject Code : BSB02

Time Allowed : 3 Hours

Full Marks : 70

*Candidate are required to give the answers in their own words
as far as practicable.*

Answer five questions in which Question No. 1 is compulsory.

All questions are of equal value.

1. Choose the correct answer:

2×7=14

(i) Which of the following statements is false?

- (a) The reaction tends to go in the forward direction if ΔG is large and positive.
- (b) The reaction moves in the backward direction if ΔG is large and negative.
- (c) The system is at equilibrium if ΔG is 0.
- (d) The reaction tends to move in the backward direction if ΔG is large and positive.

(ii) Which of the following enzyme catalyzes the first step of glycolysis?

- (a) Hexokinase
- (b) Pyruvate kinase
- (c) Glucokinase
- (d) Phosphofructokinase-1

(iii) Which of the following statements does not accurately describe synapses?

- (a) Chemical synapses have a gap between the neuron and its target cell.
- (b) Electrical synapses have no gaps between the neuron and its target cells.
- (c) Both chemical and electrical synapses relay information from neurons to target cells.
- (d) Both chemical and electrical synapses relay information through similar mechanisms.

(iv) Which one of the following components was not a part of Miller and Urey experiment?

- (a) Hydrogen
- (b) Methane
- (c) Oxygen
- (d) Ammonia

(v) The main difference between animal and plant cell is

- (a) absence of cell wall in animal cells.
- (b) absence of endoplasmic reticulum in animal cells.
- (c) absence of cell membrane in plant cells.
- (d) All of the above

- (vi) Which of the following is not a polymer?
- (a) Starch (b) Nucleic Acid
(c) Fructose (d) Protein
- (vii) Which of the following statements is true about metabolism?
- (a) It only involves the processes that build up the large molecules in the body.
(b) It only involves the processes that break down the small molecules in the body.
(c) It only involves the processes that break down the large molecules in the body.
(d) It involves all of the processes that break down and build up molecules in the body.
2. (a) What is an artificial pacemaker? Differentiate between R-wave inhibited and R-wave triggered pacemaker.
(b) Describe the instrumentation in obtaining the brain signals. 7+7
3. (a) Classify different lipids with their role in the biological system.
(b) Explain the organization of life from cell to the organism. 7+7
4. (a) Compare the filtration method in the kidney to the artificial kidney.
(b) Compare the working of the CPU to the brain. 7+7
5. (a) Explain the mechanism of operation of smart stick for the visually impaired.
(b) What is biomaterial? Explain different types of biomaterials with biological applications. 7+7
6. (a) Explain the principle of biosensors using a suitable example.
(b) What is bioimaging? Elaborate on the application of X-ray machine to image the body. 7+7
7. (a) Explain the role of computation techniques in understanding the biological system.
(b) Explain the application of AI techniques in healthcare system. 7+7
-